# Formulae, calculations and ionic bonding

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

## 1 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [1]

**Answer: B**

### Worked solution

The subscript 2 after the bracket multiplies everything inside the bracket: 2 N and 8 H. So Mr = (2 × 14) + (8 × 1) + 32 + (4 × 16) = 28 + 8 + 32 + 64 = 132.

## 2 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [1]

**Answer: D**

### Worked solution

Work out the Mr of each substance first, remembering that oxygen gas is $O_2$ (Mr 32), then divide each mass by its Mr. Water gives 0.50 mol, the greatest value.

## 3 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [1]

**Answer: A**

### Worked solution

The empirical formula is the simplest whole-number ratio of atoms. Find the largest number that divides 6, 12 and 2 exactly (2) and divide each by it: $C_3H_6O$.

## 4 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [1]

**Answer: C**

### Worked solution

Convert the volume: 250 $cm^3$ = 0.250 $dm^3$. Amount of NaOH = concentration × volume = 0.200 × 0.250 = 0.0500 mol. Mr of NaOH = 23 + 16 + 1 = 40, so mass = 0.0500 × 40 = 2.00 g.

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## 5 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [1]

**Answer: A**

### Worked solution

n(Mg) = 0.36 ÷ 24 = 0.015 mol. From the equation, Mg : $H_2$ = 1 : 1, so n($H_2$) = 0.015 mol. Volume = 0.015 × 24 $dm^3$ = 0.36 $dm^3$ = 360 $cm^3$.

## 6 · Part 2

Paper 1 & 2 | Ionic bonding | [1]

**Answer: C**

### Worked solution

Find the lowest common multiple of the charges 3 and 2, which is 6. Two $Al^{3+}$ ions give 6+ and three $SO_4^{2-}$ ions give 6–. Brackets are needed because there is more than one sulfate ion.

## 7 · Part 2

Paper 1 & 2 | Ionic bonding | [1]

**Answer: D**

### Worked solution

Losing two negative electrons leaves the atom with two more protons than electrons, so the ion has a charge of 2+: $Mg^{2+}$.

## 8 · Part 2

Paper 1 & 2 | Ionic bonding | [1]

**Answer: B**

### Explanation

An ionic bond is the strong electrostatic attraction between oppositely charged ions. A describes a covalent bond, C describes metallic bonding, and D describes forces between molecules, not ionic bonding.

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## 9 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

**Mark scheme**

**(a)** **[2]**

- calcium chloride (1)
- water and carbon dioxide (1)

**ALLOW** the reactants calcium carbonate + hydrochloric acid on the left with the three products on the right in any order

**REJECT** formulae for (a) unless every name is also correct

**(b)** **[2]**

- 2 in front of HCl (1)
- state symbols: $CaCl_2(aq)$, $H_2O(l)$, $CO_2(g)$, all three correct (1)

**(c)** **[2]**

- $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$, correct formulae and balanced (1)
- state symbols: $Al(s)$, $Fe_2O_3(s)$, $Al_2O_3(s)$, $Fe(l)$ (1)

**ALLOW** the state symbol mark for correct state symbols on correct formulae even if the equation is not balanced

**ALLOW** multiples, e.g. $4Al + 2Fe_2O_3 \rightarrow 2Al_2O_3 + 4Fe$

**Worked solution**

(a) An acid and a carbonate react to give a salt, water and carbon dioxide: calcium carbonate + hydrochloric acid → calcium chloride + water + carbon dioxide.

(b) $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$. There are 2 Cl in $CaCl_2$, so 2HCl is needed; this also gives the 2 H in $H_2O$. Calcium chloride is dissolved (aq), water is a liquid (l) and carbon dioxide is a gas (g).

(c) Aluminium forms $Al^{3+}$ ions and oxide ions are $O^{2-}$, so aluminium oxide is $Al_2O_3$. $Fe_2O_3$ has 2 Fe and 3 O; $Al_2O_3$ has 2 Al and 3 O. So: $2Al(s) + Fe_2O_3(s) \rightarrow Al_2O_3(s) + 2Fe(l)$. Check: Al 2 = 2, Fe 2 = 2, O 3 = 3.

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## 10 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

**Mark scheme**

**(a)** **[1]**

- mole OR mol (1)

**(b)** **[1]**

- $(2 \times 14) + (4 \times 1) + (3 \times 16) = 28 + 4 + 48 = 80$ shown (1)

**(c)** **[2]**

- 2.0 kg = 2000 g (1)
- $2000 \div 80 = 25$ mol (1)

**ALLOW** 0.025 kmol for both marks

- 0.025 mol (no conversion to grams) scores 1 mark only

**(d)** **[2]**

- Mr of urea = $12 + 16 + 2 \times (14 + 2) = 60$ (1)
- mass = $0.40 \times 60 = 24$ g (1)

**ECF** from an incorrect Mr

**Worked solution**

(a) The amount of a substance is measured in moles (mol).

(b) $NH_4NO_3$ contains 2 N, 4 H and 3 O: 28 + 4 + 48 = 80.

(c) Change kilograms to grams first: 2.0 kg = 2000 g. Amount = mass ÷ Mr = 2000 ÷ 80 = 25 mol.

(d) $CO(NH_2)_2$ contains 1 C, 1 O, 2 N and 4 H: 12 + 16 + 28 + 4 = 60. Mass = amount × Mr = 0.40 × 60 = 24 g.

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## 11 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

**Mark scheme**

**(a)** **[1]**

- $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$ (1)

**(b)** **[3]**

- $n(Na) = 4.6 \div 23 = 0.20$ mol (1)
- $n(NaCl) = 0.20$ mol (1)
- mass $= 0.20 \times 58.5 = 11.7$ g (1)

**ALLOW** 12 g

**ECF** from an incorrect mole ratio

**(c)** **[2]**

- $n(Cl_2) = 0.20 \div 2 = 0.10$ mol (1)
- mass $= 0.10 \times 71 = 7.1$ g (1)

**ALLOW** 11.7 − 4.6 = 7.1 g (conservation of mass, using the answer to (b)) for both marks

**ECF** from (b)

**Worked solution**

(a) Each $Cl_2$ molecule gives two chloride ions, so 2NaCl form, which needs 2Na.

(b) $n(Na) = 4.6 \div 23 = 0.20$ mol. The ratio Na : NaCl is 2 : 2 = 1 : 1, so 0.20 mol NaCl forms. Mr of NaCl = 23 + 35.5 = 58.5, so mass $= 0.20 \times 58.5 = 11.7$ g.

(c) The ratio Na : $Cl_2$ is 2 : 1, so $n(Cl_2) = 0.10$ mol. Mr of $Cl_2$ = 71, so mass $= 0.10 \times 71 = 7.1$ g. Check using conservation of mass: 4.6 g + 7.1 g = 11.7 g, which matches (b).

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## 12 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [7]

**Mark scheme**

**(a)** **[3]**

- Mr of $CaCO_3$ = 100 and Mr of CaO = 56 (1)
- n($CaCO_3$) = 50 000 ÷ 100 = 500 mol (1)
- mass of CaO = 500 × 56 = 28 000 g = 28.0 kg (1)

**ALLOW** ratio method: 50.0 × 56 ÷ 100 = 28.0 kg for full marks

**ALLOW** 28 kg

**(b)** **[2]**

- (25.2 ÷ 28.0) × 100 (1)
- 90.0% (1)

**ALLOW** 90%

**ECF** from (a)

**(c)** **[2]**

*Any two from:*

- not all the calcium carbonate decomposed OR heating was not long enough OR temperature not high enough (1)
- some solid (calcium oxide) was lost, e.g. as dust or when it was removed from the kiln (1)
- the calcium carbonate (limestone) was impure (1)

**IGNORE** "human error" or "measurement error" without detail

**Worked solution**

(a) Mr: $CaCO_3$ = 40 + 12 + 48 = 100; CaO = 40 + 16 = 56. 50.0 kg = 50 000 g, so n($CaCO_3$) = 50 000 ÷ 100 = 500 mol. The ratio $CaCO_3$ : CaO is 1 : 1, so 500 mol CaO forms: 500 × 56 = 28 000 g = 28.0 kg. Check: 100 g of $CaCO_3$ gives 56 g of CaO, so 50.0 kg gives 50.0 × 0.56 = 28.0 kg.

(b) Percentage yield = (actual yield ÷ theoretical yield) × 100 = (25.2 ÷ 28.0) × 100 = 90.0%.

(c) The yield is lower if some calcium carbonate does not decompose, if some product is lost, or if the starting material is not pure calcium carbonate (so the theoretical yield calculated from 50.0 kg is too high).

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## 13 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

**Mark scheme**

**(a)** **[1]**

- the simplest whole-number ratio of atoms of each element in a compound (1)

**(b)** **[3]**

- C 54.5 ÷ 12 = 4.54; H 9.1 ÷ 1 = 9.1; O 36.4 ÷ 16 = 2.275 (1)
- divide by the smallest (2.275): C 2.0, H 4.0, O 1 (1)
- $C_2H_4O$ (1)

**ALLOW** atoms in any order, e.g. $C_2OH_4$

**(c)** **[2]**

- Mr of $C_2H_4O$ = 44 (1)
- 88 ÷ 44 = 2, so $C_4H_8O_2$ (1)

**ECF** from an incorrect empirical formula in (b)

**Worked solution**

(a) The empirical formula shows the simplest whole-number ratio of atoms of each element in the compound.

(b) Imagine 100 g of X: it contains 54.5 g C, 9.1 g H and 36.4 g O. Divide by Ar: C 54.5 ÷ 12 = 4.54, H 9.1 ÷ 1 = 9.1, O 36.4 ÷ 16 = 2.275. Divide each by the smallest value, 2.275: C 2.00, H 4.00, O 1.00. The empirical formula is $C_2H_4O$.

(c) Mr of $C_2H_4O$ = 24 + 4 + 16 = 44. The molecular formula has 88 ÷ 44 = 2 empirical units, so it is $C_4H_8O_2$. Check: (4 × 12) + (8 × 1) + (2 × 16) = 48 + 8 + 32 = 88.

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## 14 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [7]

**Mark scheme**

**(a)** **[2]**

- to make sure all the water of crystallisation has been removed (1)
- constant mass shows that no more water is being lost OR the reaction is complete (1)

**(b)** **[2]**

- mass of anhydrous magnesium sulfate = 27.70 − 25.30 = 2.40 g (1)
- mass of water = 30.22 − 27.70 = 2.52 g (1)

**REJECT** 28.05 or 27.72 as the final mass

**(c)** **[3]**

- $n(MgSO_4) = 2.40 \div 120 = 0.0200$ mol (1)
- $n(H_2O) = 2.52 \div 18 = 0.140$ mol (1)
- $x = 0.140 \div 0.0200 = 7$ (1)

**ECF** from (b)

**Worked solution**

(a) Each heating drives off more water. When two weighings in a row give the same mass (27.70 g), no more water is leaving, so the solid is fully anhydrous. Stopping earlier would leave some water behind and give a value of x that is too small.

(b) Use the final constant mass. Anhydrous $MgSO_4$ = 27.70 − 25.30 = 2.40 g. Water lost = 30.22 − 27.70 = 2.52 g.

(c) Mr of $MgSO_4$ = 24 + 32 + (4 × 16) = 120; Mr of $H_2O$ = 18. $n(MgSO_4) = 2.40 \div 120 = 0.0200$ mol; $n(H_2O) = 2.52 \div 18 = 0.140$ mol. Ratio = 0.140 ÷ 0.0200 = 7, so the formula is $MgSO_4 \cdot 7H_2O$. Check: Mr of $MgSO_4 \cdot 7H_2O$ = 120 + 126 = 246, and 0.0200 × 246 = 4.92 g, which equals 30.22 − 25.30.

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## 15 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [8]

**Mark scheme**

**(a)** **[4]**

*Any four from:*

- weigh the empty crucible and lid (1)
- add the magnesium ribbon (coiled) and weigh again (1)
- heat the crucible strongly on the pipe-clay triangle and tripod (1)
- lift the lid slightly from time to time to let air in (1)
- continue heating until the magnesium no longer glows OR stops reacting (1)
- allow to cool and weigh the crucible, lid and contents (1)
- reheat, cool and reweigh until the mass is constant (1)

**ALLOW** cleaning the magnesium ribbon with sandpaper first to remove the oxide layer as a marking point

**(b)** **[3]**

- mass of Mg = 0.36 g and mass of O = 0.24 g (1)
- n(Mg) = 0.36 ÷ 24 = 0.015 AND n(O) = 0.24 ÷ 16 = 0.015 (1)
- ratio 1 : 1, so MgO (1)

**ECF** from incorrect masses

**(c)** **[1]**

- to let oxygen (air) in while stopping magnesium oxide (smoke) escaping (1)

**ALLOW** to let air in without losing any product

**ALLOW** either idea alone: to let oxygen/air reach the magnesium OR to stop magnesium oxide (white smoke/powder) escaping so that no product is lost

**Worked solution**

(a) Weigh the crucible and lid, add the magnesium and weigh again; the difference is the mass of magnesium. Heat strongly, lifting the lid a little now and then so that oxygen can reach the magnesium. When the magnesium stops glowing, cool and weigh. Heat, cool and reweigh until the mass stays the same, so all the magnesium has reacted. The final mass minus the crucible and lid gives the mass of magnesium oxide.

(b) Mass of Mg = 22.84 − 22.48 = 0.36 g. Mass of oxygen gained = 23.08 − 22.84 = 0.24 g. n(Mg) = 0.36 ÷ 24 = 0.015 mol; n(O) = 0.24 ÷ 16 = 0.015 mol. The ratio is 1 : 1, so the empirical formula is MgO.

(c) Oxygen from the air is needed for the magnesium to burn, but if the lid is left off, white magnesium oxide smoke escapes and the mass of product is too low.

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## 16 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [6]

**Mark scheme**

**(a)** **[3]**

- n(NaCl) = 5.85 ÷ 58.5 = 0.100 mol (1)
- volume = 250 ÷ 1000 = 0.250 $dm^3$ (1)
- concentration = 0.100 ÷ 0.250 = 0.400 mol/$dm^3$ (1)

**ALLOW** 0.4 mol/$dm^3$

**ECF** from an incorrect amount

**(b)** **[1]**

- 0.400 × (25.0 ÷ 1000) = 0.0100 mol (1)

**ECF** from (a)

**(c)** **[2]**

- amount of NaCl stays 0.0100 mol AND volume = 0.200 $dm^3$ (1)
- 0.0100 ÷ 0.200 = 0.0500 mol/$dm^3$ (1)

**ALLOW** 0.05 mol/$dm^3$

**ECF** from (b)

**Worked solution**

(a) Mr of NaCl = 23 + 35.5 = 58.5, so n = 5.85 ÷ 58.5 = 0.100 mol. The volume must be in $dm^3$: 250 $cm^3$ = 0.250 $dm^3$. Concentration = amount ÷ volume = 0.100 ÷ 0.250 = 0.400 mol/$dm^3$.

(b) Amount = concentration × volume = 0.400 × 0.0250 = 0.0100 mol.

(c) Adding water does not change the amount of sodium chloride, only the volume. New concentration = 0.0100 ÷ 0.200 = 0.0500 mol/$dm^3$. Check: the volume increased 8 times (25.0 to 200 $cm^3$), so the concentration falls to 0.400 ÷ 8 = 0.0500 mol/$dm^3$.

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## 17 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [7]

**Mark scheme**

**(a)** **[1]**

- $H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$ (1)

**(b)** **[1]**

- $(20.05 + 19.95 + 20.00) \div 3 = 20.00\ cm^3$ (1)

**ALLOW** $20\ cm^3$

**(c)** **[4]**

- $n(H_2SO_4) = 0.100 \times (20.00 \div 1000) = 0.00200$ mol (1)
- $n(NaOH) = 2 \times 0.00200 = 0.00400$ mol (1)
- volume of NaOH $= 25.0 \div 1000 = 0.0250\ dm^3$ (1)
- concentration $= 0.00400 \div 0.0250 = 0.160\ mol/dm^3$ (1)

**ECF** from (b) and from an incorrect mole ratio

**ALLOW** $0.16\ mol/dm^3$

**(d)** **[1]**

- it is only approximate OR the end-point was overshot OR it is not close to (concordant with) the other results (1)

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**Worked solution**

(a) Each $H_2SO_4$ provides two $H^+$ ions, so it needs 2NaOH, and 2 water molecules form.

(b) Leave out the rough value ($20.80\ cm^3$) and average the other three: $(20.05 + 19.95 + 20.00) \div 3 = 20.00\ cm^3$.

(c) $n(H_2SO_4) = 0.100 \times 0.02000 = 0.00200$ mol. From the equation, 1 mol $H_2SO_4$ reacts with 2 mol NaOH, so $n(NaOH) = 0.00400$ mol. This was in $25.0\ cm^3 = 0.0250\ dm^3$, so concentration $= 0.00400 \div 0.0250 = 0.160\ mol/dm^3$. Check: $0.160 \times 0.0250 = 0.00400$ mol NaOH, which is twice 0.00200 mol $H_2SO_4$.

(d) The rough titration is done quickly to find the approximate end-point, so the acid is usually added past the end-point. Its value ($20.80\ cm^3$) is clearly higher than the others.

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## 18 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [7]

**Mark scheme**

**(a)** **[3]**

- $n(H_2O_2) = 0.68 \div 34 = 0.020$ mol (1)
- $n(O_2) = 0.020 \div 2 = 0.010$ mol (1)
- volume = $0.010 \times 24\,000 = 240$ $cm^3$ (1)

**ECF** from an incorrect amount or ratio

**(b)** **[2]**

- $n(O_2) = 900 \div 24\,000 = 0.0375$ mol (1)
- mass = $0.0375 \times 32 = 1.2$ g (1)

**ALLOW** 1.20 g

**(c)** **[2]**

- equal volumes of gases at the same temperature and pressure (rtp) contain the same amount in moles (1)
- the Mr of carbon dioxide (44) is greater than the Mr of oxygen (32) (1)

**Worked solution**

(a) Mr of $H_2O_2 = 2 + 32 = 34$, so $n(H_2O_2) = 0.68 \div 34 = 0.020$ mol. The ratio $H_2O_2 : O_2$ is 2 : 1, so $n(O_2) = 0.010$ mol. Volume = $0.010 \times 24\,000$ $cm^3 = 240$ $cm^3$.

(b) $n(O_2) = 900 \div 24\,000 = 0.0375$ mol. Mr of $O_2 = 32$, so mass = $0.0375 \times 32 = 1.2$ g.

(c) At rtp, 1 mol of any gas occupies 24 $dm^3$, so 100 $cm^3$ of each gas contains the same amount in moles ($100 \div 24\,000 = 0.00417$ mol). Each mole of $CO_2$ has a mass of 44 g but each mole of $O_2$ has a mass of only 32 g, so the carbon dioxide sample is heavier.

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## 19 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [8]

**Mark scheme**

**(a)** **[1]**

- 1.95 ÷ 65 = 0.0300 mol (1)

**(b)** **[1]**

- 0.500 × (50.0 ÷ 1000) = 0.0250 mol (1)

**(c)** **[2]**

- zinc and sulfuric acid react in a 1 : 1 ratio (1)
- there is more zinc (0.0300 mol) than acid (0.0250 mol), so zinc is in excess (1)

**ECF** from (a) and (b)

**(d)** **[2]**

- $n(H_2)$ = 0.0250 mol (from the acid, the limiting reactant) (1)
- volume = 0.0250 × 24 000 = 600 $cm^3$ (1)

**ECF** from (c)

**REJECT** 720 $cm^3$ (calculated from zinc) for the second mark unless (c) concluded acid in excess

**(e)** **[2]**

- n(Zn) left = 0.0300 − 0.0250 = 0.0050 mol (1)
- mass = 0.0050 × 65 = 0.325 g (1)

**ALLOW** 0.33 g OR 0.3 g

**ECF** from (a), (b) and (c)

**Worked solution**

(a) n(Zn) = 1.95 ÷ 65 = 0.0300 mol.

(b) 50.0 $cm^3$ = 0.0500 $dm^3$, so $n(H_2SO_4)$ = 0.500 × 0.0500 = 0.0250 mol.

(c) The equation shows 1 mol Zn reacts with 1 mol $H_2SO_4$. Only 0.0250 mol of acid is available, so only 0.0250 mol of zinc can react. Zinc is in excess and the acid is used up.

(d) The limiting reactant (the acid) decides how much hydrogen forms. $H_2SO_4$ : $H_2$ = 1 : 1, so $n(H_2)$ = 0.0250 mol. Volume = 0.0250 × 24 000 = 600 $cm^3$.

(e) Zinc left = 0.0300 − 0.0250 = 0.0050 mol. Mass = 0.0050 × 65 = 0.325 g. Check: zinc that reacted = 0.0250 × 65 = 1.625 g, and 1.95 − 1.625 = 0.325 g.

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## 20 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [7]

**Mark scheme**

**(a)** **[3]**

- mass of Cu = 32.69 − 30.15 = 2.54 g AND mass of O = 33.33 − 32.69 = 0.64 g (1)
- n(Cu) = 2.54 ÷ 63.5 = 0.0400 AND n(O) = 0.64 ÷ 16 = 0.0400 (1)
- ratio 1 : 1, so CuO (1)

**ECF** from incorrect masses

**(b)** **[2]**

- to stop air (oxygen) reaching the hot copper (1)
- otherwise the copper would be oxidised again (react with oxygen), making the final mass too high (1)

**(c)** **[2]**

- $n(H_2) = n(O) = 0.0400$ mol (1)
- volume = 0.0400 × 24 000 = 960 $cm^3$ (1)

**ECF** from the amount of oxygen in (a)

**ALLOW** 0.96 $dm^3$ for the second mark only if the unit is given as $dm^3$

**Worked solution**

(a) Mass of oxide = 33.33 − 30.15 = 3.18 g. Mass of copper = 32.69 − 30.15 = 2.54 g. Mass of oxygen removed = 3.18 − 2.54 = 0.64 g. n(Cu) = 2.54 ÷ 63.5 = 0.0400 mol; n(O) = 0.64 ÷ 16 = 0.0400 mol. The ratio is 1 : 1, so the empirical formula is CuO. Check: Mr of CuO = 79.5 and 3.18 ÷ 79.5 = 0.0400 mol.

(b) Hot copper reacts easily with oxygen in air. Keeping hydrogen flowing keeps air out until the copper is cool, so none of it turns back into the oxide.

(c) 0.0400 mol of oxygen atoms is removed, and each needs one $H_2$ molecule, so $n(H_2) = 0.0400$ mol. Volume = 0.0400 × 24 000 = 960 $cm^3$. In practice more hydrogen than this is used, because some passes through without reacting.

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## 21 · Part 2 Paper 1 & 2 Ionic bonding [6]

**Mark scheme**

**(a)** **[2]**

- calcium ion 2+ ($Ca^{2+}$) (1)
- chloride ion 1– ($Cl^-$) (1)

**(b)** **[3]**

- each calcium atom loses (its) two outer electrons (1)
- each chlorine atom gains one electron (1)
- electrons are transferred from calcium to chlorine, one calcium atom to two chlorine atoms (1)

**ALLOW** the ions have a full outer shell OR noble gas electronic configuration (2.8.8) as an alternative third mark

**REJECT** sharing of electrons for any mark

**(c)** **[1]**

- $CaCl_2$ (1)

**Worked solution**

(a) Calcium (2.8.8.2) has two outer electrons, which it loses to form $Ca^{2+}$. Chlorine (2.8.7) needs one more electron to fill its outer shell, so it forms $Cl^-$.

(b) Each calcium atom transfers its two outer electrons, one to each of two chlorine atoms. The calcium atom becomes $Ca^{2+}$ (2.8.8) and each chlorine atom becomes $Cl^-$ (2.8.8). Both ions have full outer shells.

(c) One $Ca^{2+}$ balances two $Cl^-$, so the formula is $CaCl_2$.

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## 22 · Part 2 Paper 1 & 2 Ionic bonding [6]

**Mark scheme**

**(a)** **[4]**

- potassium oxide: $K_2O$ (1)
- iron(III) hydroxide: $Fe(OH)_3$ (1)
- ammonium carbonate: $(NH_4)_2CO_3$ (1)
- zinc nitrate: $Zn(NO_3)_2$ (1)

**REJECT** $FeOH_3$, $NH_{42}CO_3$ and $ZnNO_{32}$ (missing brackets)

**(b)** **[2]**

- calcium forms $Ca^{2+}$ and nitrogen (Group 5) forms $N^{3-}$ (1)
- three $Ca^{2+}$ give 6+ and two $N^{3-}$ give 6–, so the charges balance (compound is neutral) (1)

**Worked solution**

(a) Combine the ions so the total charge is zero. $K^+$ and $O^{2-}$: two $K^+$ for one $O^{2-}$, $K_2O$. $Fe^{3+}$ and $OH^-$: three $OH^-$ needed, $Fe(OH)_3$. $NH_4^+$ and $CO_3^{2-}$: two $NH_4^+$ needed, $(NH_4)_2CO_3$. $Zn^{2+}$ and $NO_3^-$: two $NO_3^-$ needed, $Zn(NO_3)_2$. Brackets are used when there is more than one of a group of atoms.

(b) Calcium is in Group 2, so it forms $Ca^{2+}$. Nitrogen atoms gain three electrons to form $N^{3-}$. The lowest common multiple of 2 and 3 is 6: 3 × (2+) = 6+ and 2 × (3–) = 6–, so the formula is $Ca_3N_2$.

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## 23 · Part 2 Paper 1 & 2 Ionic bonding [6]

**Mark scheme**

**(a)** **[4]**

- (sodium chloride has a) giant (ionic) lattice OR giant structure (1)
- made of oppositely charged ions OR $Na^+$ and $Cl^-$ ions (1)
- strong electrostatic attraction between the ions (1)
- a lot of (heat) energy is needed to overcome the attractions OR there are many attractions to overcome (1)

**REJECT** reference to molecules, intermolecular forces or covalent bonds for the relevant mark

**(b)** **[2]**

- magnesium and oxide ions ($Mg^{2+}$ and $O^{2-}$) have larger charges than sodium and chloride ions ($Na^+$ and $Cl^-$) (1)
- so the electrostatic attraction between the ions is stronger and more energy is needed to overcome it (1)

**ALLOW** $Mg^{2+}$ and $O^{2-}$ are smaller than $Na^+$ and $Cl^-$ (ions closer together) as an alternative to larger charges for the first mark

**Worked solution**

(a) Sodium chloride is a giant lattice of positive sodium ions and negative chloride ions. Every ion is held in place by strong electrostatic attractions to the oppositely charged ions around it, in all directions. To melt the solid, many of these strong attractions must be overcome, which needs a lot of energy, so the melting point is high.

(b) In magnesium oxide the ions are $Mg^{2+}$ and $O^{2-}$, with twice the charge of $Na^+$ and $Cl^-$. Larger charges mean stronger electrostatic attraction between the ions, so even more energy is needed to overcome the attractions and the melting point is much higher.

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## 24 · Part 2 Paper 1 & 2 Ionic bonding [6]

**Mark scheme**

**(a)** **[4]**

- zinc chloride contains ions (1)
- in the solid, the ions are held in fixed positions (in the lattice) and cannot move (1)
- when molten or dissolved in water, the ions are free to move (1)
- moving ions carry the charge (current) through the liquid (1)

**REJECT** electrons moving through the compound for the last mark

**(b)** **[1]**

- zinc chloride has a much lower melting point, so it can be melted using a Bunsen burner (school laboratory equipment) (1)

**ALLOW** sodium chloride's melting point is too high to reach in a school laboratory

**(c)** **[1]**

- $ZnCl_2$ (1)

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**Worked solution**

(a) For a substance to conduct, charged particles must be able to move. Zinc chloride is made of $Zn^{2+}$ and $Cl^-$ ions. In the solid these ions are fixed in the lattice, so no charge flows and the lamp stays off. Melting or dissolving breaks up the lattice, so the ions can move; the moving ions carry the charge and the lamp lights.

(b) A Bunsen burner can heat zinc chloride above 290 °C, but sodium chloride needs over 800 °C, which is very difficult to reach safely in a school laboratory.

(c) Zinc ions are $Zn^{2+}$ and chloride ions are $Cl^-$, so two chloride ions balance one zinc ion: $ZnCl_2$.

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## 25 · Part 2 Paper 1 & 2 Ionic bonding [7]

**Mark scheme**

**(a)** **[2]**

- $Rb^+$ and $Se^{2-}$ (1)
- $Rb_2Se$ (1)

**ECF** for the formula from incorrect ion charges

**(b)** **[3]**

- K (1)
- K does not conduct when solid but does conduct when molten (1)
- K has a high melting point (1)

**ALLOW** J is not ionic because it conducts when solid OR L is not ionic because it has a low melting point and does not conduct when molten, as the explanation for one mark only if the identification is K

- No explanation marks if the substance identified is not K

**(c)** **[2]**

- a rubidium atom has one outer electron, which it loses (to get a full outer shell) (1)
- a selenium atom has six outer electrons and gains two electrons (to get a full outer shell) (1)

**Worked solution**

(a) Group 1 metals lose one electron to form 1+ ions, so rubidium forms $Rb^+$. Group 6 non-metals gain two electrons to form 2– ions, so selenium forms $Se^{2-}$. Two $Rb^+$ balance one $Se^{2-}$, so the formula is $Rb_2Se$.

(b) Ionic compounds do not conduct when solid, because the ions cannot move, but do conduct when molten, because the ions can move. They also have high melting points because of the strong attractions between ions. K fits all of these. J conducts when solid, so it behaves like a metal. L has a low melting point and does not conduct even when molten, so it contains no ions.

(c) Rubidium is in Group 1, so its atom has one outer electron; losing it leaves a full outer shell and a charge of 1+. Selenium is in Group 6, so its atom has six outer electrons; gaining two electrons fills the outer shell and gives a charge of 2–.

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